Applied Machine Learning (CSAI2017P)

Student Course Handbook

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B.Tech CSE (AI & ML), Semester III • Autumn 2026

What this handbook is

Everything you need to know about how this course runs: what you will learn, how you will be assessed, when each thing is due, and where to get help.
Keep it. Almost every question you are likely to ask this semester is answered somewhere in these pages. Anything that changes will be announced on the LMS and this document reissued.

Contents

1 The course at a glance

Course title	Applied Machine Learning
Course code	CSAI2017 (theory) • **CSAI2017P** (theory + lab)
Credits (L-T-P-C)	4-0-1-5
Programme	B.Tech CSE (Artificial Intelligence & Machine Learning)
Semester	III
Session	Autumn 2026 (Aug–Dec), AY 2026-27
Units	7
Theory	24 lecture packages across 40 sessions
Laboratory	15 experiments
Prerequisites	Elements of AI & ML (CSAI2018); Python Programming (CSEG1021)
Faculty	Dr. Tofik Ali
Repository	`https://github.com/tali7c/Applied-Machine-Learning`

1.1 Course outcomes

By the end of the course you should be able to:

- **CO1** — Understand the core concepts and techniques of machine learning and artificial intelligence.
- **CO2** — Develop machine learning models using popular libraries and frameworks.
- **CO3** — Evaluate the performance of machine learning models using appropriate metrics.
- **CO4** — Apply machine learning to various real-world problems and domains.

1.2 What you will actually be able to do

By the end of the semester you will have built, from scratch, a housing-price predictor, a stock forecaster, a churn model, a digit recogniser, a spam filter, a credit scorer, an anomaly detector, a breast-cancer diagnostic model and a crop-disease classifier — and, more importantly, you will be able to tell when one of them is lying to you.

2 Syllabus

The order is deliberate. Units II and III come early because once you know **what quantity we are minimising** and **how the minimisation is carried out**, every method in Units V–VII is a variation on one theme rather than a new thing to memorise.

	Unit	Sessions	Content
I	Introduction	2	Overview of machine learning and its applications. Types of machine learning: supervised, unsupervised, reinforcement. Python and its libraries for ML (NumPy, pandas, scikit-learn).
II	Loss Functions	2	Mean Squared Error, Mean Absolute Error, Huber loss. Binary, Categorical and Sparse Categorical Cross-Entropy. Hinge loss (SVM loss), Triplet loss.
III	Optimizer Functions	4	Stochastic gradient descent. Mini-batch gradient descent. Momentum. Adagrad. Adam. RMSprop. Adadelta. Numerical practice.
IV	Data Preprocessing	6	Data cleaning; handling missing data and outliers; data transformation. Feature engineering, scaling and encoding. Data reduction and dimensionality reduction; feature selection. Train–test splitting and cross-validation. Handling imbalanced data.
V	Regression	7	Introduction to regression. Simple linear regression and its proof. Least-squares estimation; line of best fit. Maximum likelihood estimation. R^2; model adequacy; over-fitting and its detection. Logistic regression. Multiple linear regression and coefficient interpretation. Ridge and lasso regularization.
VI	Classification	12	ML classifiers and instance-based learning. K-Nearest Neighbour. Decision trees; information gain; ID3. Bayesian algorithms. Ensembles: bagging, boosting, random forests. Neural networks; activation functions; multi-layer perceptron; backpropagation. Support vector machines. Model evaluation; ROC and AUC.
VII	Clustering Techniques	7	Introduction to clustering; cluster statistics and applications. Similarity and dissimilarity; data types in clustering. Hierarchical clustering and HAC; linkage methods; cluster distance measures. K-means; K-medoids; CLARA. Density-based methods; DBSCAN.

Total: **40 theory sessions**, including two assessment sessions and a final revision session.

About the session counts

The unit contents above are the official syllabus. The *per-unit session counts* are this course's planned allocation of the available contact sessions, not a figure from the Academic Handbook, and they may shift by a session or two as the semester runs. Treat them as a guide to the relative weight of each unit, not as a contract.

3 How you are assessed

Indicative

The component split below is subject to departmental approval. The confirmed scheme will be posted on the LMS, and this handbook reissued if it changes. The Internal 50 : Mid-Semester 20 : End-Semester 30 frame is likewise to be confirmed against the Academic Handbook.

3.1 The 100 marks

Component	Marks	What it is
Theory assignments	5	Two assignments, each followed by a short in-class assessment test
Theory quizzes / tests	10	Ten 40-minute tests through the semester
Project — individual	10	A quiz on your own group's project; **marked individually**
Project — group	5	The state of what your group submitted
Lab assignments	10	Ten assignments covering experiments 1–14
Lab quizzes	10	Ten 60-minute quizzes on *your own* submitted lab code
Mid-Semester examination	20	Common paper, university schedule
End-Semester examination	30	Common paper, university schedule
Total	**100**	

3.2 How that maps to the university scheme

Block	Marks	Made up of
Continuous Internal Assessment (CIA)	50	everything except the two examinations
Mid-Semester Examination	20	
End-Semester Examination	30	

The single most important line in this handbook

Seventy of your hundred marks are decided before the end-semester examination — fifty from work spread across the whole semester (quizzes, assignments, lab work, project) and twenty at the mid-semester. A student who coasts until the last month cannot recover, because the marks are already gone.

4 The assessments in detail

4.1 Theory quizzes — 10 marks

Ten tests, T1–T10, each **40 minutes**. The diagnostic Test-0 is shorter, at **30 minutes**.

Test	Coverage
T0	Prerequisites and general understanding — **no marks**
T1	Units I–II
T2	Unit III; Unit IV (cleaning)
T3	Unit IV
T4	Unit V — foundations, least squares, MLE
T5	Unit V — logistic, MLR, regularization
T6	Unit VI — foundations, KNN, decision trees
T7	Unit VI — Bayesian, ensembles
T8	Unit VI — neural networks, SVM, evaluation
T9	Unit VII — foundations, similarity, data types
T10	Unit VII — hierarchical, partitioning, density-based

Dates are not fixed yet

The tests run in this order and cover this material, but **the dates are not settled**. Each one will be announced at least one session in advance, on the LMS and in class. Do not plan around a date you have inferred from the syllabus — wait for the announcement.

Each test is marked out of 10; the component is the total scaled to 10. **An absence counts as zero**, so tell me in advance if you cannot sit one.

A **drop-lowest-two** provision may be applied, in which case your best 8 of 10 count. Whether it is in force will be confirmed early in the semester — plan on all ten counting and treat it as a safety net, not a budget.

About Test-0, and about the different question sets

Test-0 is a diagnostic. It carries **no marks**, is not a ranking, and is not an ability test. There is nothing to revise for it. Its purpose is to let the teaching team see *how you approach a problem*, so that the short tests through the semester can be framed in a way that suits how you work. Show your working; an incomplete answer that makes its reasoning visible is more useful than a bare correct one.

You will notice that the quizzes come in several versions. What you should know about that:

- The versions match the **style** of a question to how you work. They are not ability tiers.
- Every version asks **equally demanding** questions, on the same topics, for the same marks, and is marked against the **same rubric**.
- Which version you get may change during the semester. That is normal and is not a promotion or a demotion.
- It has **no effect whatsoever** on any mark you receive.
- T1 may run as a single common paper for everyone; the versions begin at T2 in that case.

If any of that does not match your experience, tell me — that would be a fault in the design and I would want to fix it.

4.2 Theory assignments — 5 marks

Two assignments, each worth 2.5. Each has two parts: the submission itself (1.0) and a short **in-class assessment test** on what you submitted (1.5).

Asst	Coverage	Issued
A1	Units I–IV — preprocessing and cross-validation pipeline	once Unit IV is complete
A2	Units V–VI — classifier comparison with ROC/AUC	once Unit VI is complete

Each is issued with about a fortnight to work on it; the submission date and the assessment test are announced when the assignment goes out.

Why the test is worth more than the submission

Because it is the part that proves the work is yours. If you did the assignment, the test is straightforward. If you did not, no amount of polish on the submission will save it.

4.3 Laboratory — 20 marks

Experiments 1–14 are grouped into **ten assignments**. Experiment 15, the mini-project, is assessed under the project component instead.

Asst	Expts	Topic	Quiz
LA1	1, 2	Environment setup; preprocessing and EDA	LQ1
LA2	3	Regression: housing price prediction	LQ2
LA3	4	Time-series regression: stock prices	LQ3
LA4	5	Customer churn prediction	LQ4
LA5	6, 7	Image and multiclass classification: digits and Iris	LQ5
LA6	8	Spam email detection	LQ6
LA7	9	Credit risk assessment	LQ7
LA8	10, 11	Anomaly detection; physiological signal classification	LQ8
LA9	12	Breast cancer diagnosis from imaging features	LQ9
LA10	13, 14	Clustering medical images; crop disease classification	LQ10

Each assignment falls due shortly after the experiments it covers, and its quiz follows in a later lab. **Submission deadlines and quiz dates are announced in the lab and on the LMS** — they are not fixed in advance here.

How a lab assignment is marked

Each is out of 10, with a choice structure:

Section	Level	Choice rule	Marks
A	Easy	attempt any 2 of 3 (2 marks each)	4
B	Medium	attempt any 1 of 2 (4 marks each)	4
C	Consolidation	compulsory	2
Total			**10**

Attempting more than the rule allows is not penalised — the best attempts are marked.

Lab quizzes — and why you should write your own code

Each lab quiz runs for **60 minutes at the machine** and asks you to **read, trace, debug and modify your own submitted code**. Not recall, not theory — your code, in front of you, with questions about it.

Caution

This is deliberate, and it is the reason copying does not work in this course. An hour is long enough that code you did not write becomes very hard to defend. Struggling through your own imperfect solution will score better than submitting someone else's good one.

4.4 Project — 15 marks

Ten projects are offered. You form your own group and choose one. Each project is normally taken by one group only; two groups may take the same one only with prior approval.

Sub-component	Marks	Basis
Project quiz	10	An oral and written test on your group's project. **Marked individually** — everyone is asked separately.
Project state	5	Completeness, correctness, reproducibility and documentation of the artefact. **Awarded to the group.**

#	Milestone
1	Project catalogue released
2	Groups formed and project selected
3	Checkpoint 1 — data secured, baseline model
4	Checkpoint 2 — full pipeline, initial results
5	Final submission (code, report, reproducibility notes)
6	Individual project quiz

The catalogue goes out early in the semester and the two checkpoints fall either side of the mid-semester examination. **Each date is announced when it is fixed**; the order above will not change.

Read this before you choose your group

Two thirds of the project mark is **individual**. A strong contributor in a weak group can still score well. A passenger in a strong group cannot — the group mark will not save them, because it is only 5 of the 15.
Group marks are not divided by contribution. All individual differentiation happens through the individual quiz.

5 The order of events

No dates in this handbook — and that is deliberate

The assessment calendar is not fixed yet, so publishing dates here would only create a schedule you could be misled by. What *is* fixed is the **order**, and what each item requires of you. Every date is announced on the LMS and in class as it is confirmed — a quiz at least one session ahead.

#	What happens	What you should have done
1	Unit I lectures	`pip install numpy pandas scikit-learn matplotlib seaborn jupyter`
2	**Test-0** (no marks)	Nothing — turn up
3	**LA1 due**, then **LQ1**	Experiments 1–2 written up, and be able to explain your own code
4	**T1**	Units I–II
5	Project catalogue released	Start thinking about a group
6	**T2**	Unit III; Unit IV cleaning
7	**T3**; groups and project fixed	Unit IV
8	**A1 submission + assessment test**	Assignment 1 done and understood
9	**T4**	Unit V foundations
10	**T5**; project checkpoint 1	Unit V; data secured, baseline model
11	**Mid-Semester examination**	Units I–V
12	**T6**	Unit VI foundations, KNN, trees
13	**T7**	Unit VI Bayesian, ensembles
14	**T8**; project checkpoint 2	Unit VI complete; pipeline running
15	**A2 submission + assessment test**	Assignment 2 done and understood
16	**T9**	Unit VII foundations
17	**T10**; project final submission	Unit VII complete; project handed in
18	Revision; individual project quiz	Everything
19	**End-Semester examination**	Everything

LA1–LA10 and LQ1–LQ10 run through the lab sessions in parallel with the above, roughly one assignment every one to two labs. The Mid-Semester and End-Semester examinations follow the university calendar and are notified separately by the examination section, not by me.

6 How to do well in this course

6.1 Three rules that apply to everything

1. **Fit every transformer inside a `Pipeline`.** Never fit anything on the test set — not a scaler, not an imputer, not a feature selector. This is the single most common way to produce a wonderful result that means nothing.
2. **Set `random_state` wherever reproducibility is asked for.** A result you cannot reproduce is a result you cannot defend.
3. **Every question wants a short written observation**, not just a number. Code with no commentary is capped at half marks.

6.2 Practical advice

- **Restart the kernel and run top to bottom before you submit.** Out of order cells are the commonest reason a submission does not reproduce.
- **Download each dataset once** into a local `data/` folder and keep it for the semester. `Anchor-Datasets.pdf` has the links and the loading code. Some loaders need internet on first use — run them at home, not in the lab.
- **Report metrics with units.** "MAE 0.53" means nothing; "MAE 0.53, i.e. about $53,000" means something.
- **Do not quote more precision than you have.** If cross-validation gives 0.953 ± 0.052, writing "95.33%" is false precision.
- **Ask in the lab rather than guessing.** That is what the session is for, and it costs you nothing.

6.3 What to submit

- Notebook: `AML_LAxx_<SAPID>.ipynb`, all cells executed with outputs visible.
- Report: 2–3 pages PDF, or a `README.md`, carrying your results table and observations.
- Do not upload datasets. Cite the source and include the loading code.

6.4 Academic integrity

Discussing ideas with classmates is encouraged. Submitting someone else's code or text as your own is not, and the assessment design will find it: the assignment assessment tests and the lab quizzes both ask you to explain work you have submitted. If you used a tool or a tutorial substantially, say so in your report — that is honest practice, not an admission of weakness.

7 Books, courses and other help

All links checked 4 August 2026. Free and no-login resources are marked [**free**].

Prescribed

- Müller, A. C. and Guido, S. *Introduction to Machine Learning with Python.* O'Reilly, 2016.
- Bishop, C. M. *Pattern Recognition and Machine Learning.* Springer, 2006. [**free**] from Microsoft Research.

Reference

- Murphy, K. P. *Machine Learning: A Probabilistic Perspective.* MIT Press, 2012. The author's free successor volume is at [**free**] `https://probml.github.io/pml-book/book1.html`
- Han, J., Kamber, M. and Pei, J. *Data Mining: Concepts and Techniques*, 3rd ed. Morgan Kaufmann, 2012. Especially Chapters 3, 8 and 10.
- *scikit-learn user guide* — the syllabus web resource. [**free**] `https://scikit-learn.org/stable/`

Open-access support (not on the official list, but good)

- James, G. et al. *An Introduction to Statistical Learning with Python.* [**free**] `https://www.statlearning.com/` — the friendliest treatment of regression, regularization and cross-validation available.

Courses — free, and worth your time

- **NPTEL, Introduction to Machine Learning**, Prof. Balaraman Ravindran, IIT Madras. The closest match to this syllabus. [**free**] `https://nptel.ac.in/courses/106106139`
- **SWAYAM** — national portal hosting NPTEL and other Indian university courses. [**free**] `https://swayam.gov.in/`
- **Google Machine Learning Crash Course**. [**free**] `https://developers.google.com/machine-learning/crash-course`
- **Kaggle Learn, Intro to Machine Learning** — four hours, runs in the browser, no installation. [**free**] `https://www.kaggle.com/learn/intro-to-machine-learning`

When something will not click

- **StatQuest** (Josh Starmer) — short, clear videos on almost every topic in this syllabus. Start with *A Gentle Introduction to Machine Learning.* [**free**] `https://statquest.org/video-index/`
- **MLU-Explain** — interactive visual explanations of train/test splits, bias–variance and cross-validation. [**free**] `https://mlu-explain.github.io/`
- **scikit-learn, Common Pitfalls** — the official treatment of leakage and `random_state`. Read it once, properly. [**free**] `https://scikit-learn.org/stable/common_pitfalls.html`

Course materials

Slides, notes, lab briefs and starter notebooks are published as they are delivered at `https://github.com/tali7c/Applied-Machine-Learning`

Each lecture ships with `slides.pdf` and a much fuller `notes.pdf` containing worked examples, runnable code with its real output, practice questions and full solutions. **The notes are written so you can learn the material without having been in the lecture** — use them.

Questions about anything in this handbook: ask in class, in the lab, or on the LMS.